



Simple Monophonic FM Transmitter

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June 2013

One of my earlier projects, built deadbug-style, was a simple monophonic FM transmitter based on [this design](#) - glad to see the original link still works!

The frequency can be finely tuned with a variable capacitor (finding the base frequency the first time was definitely a pain). The whole circuit lives in a metal cigar box — not my finest choice, since it interferes with the signal and only works properly if the lid's popped open.

The coil was hand-wound around a pencil, and the circuit runs off a 9 V battery. Audio input goes through a 6.3 mm to 3.5 mm jack adapter, so you can plug in almost any device. My favorite use was connecting a guitar and "playing" directly through the radio.

Notes:

- Built deadbug-style with discrete components
- Powered by a 9 V battery
- Variable capacitor for fine frequency tuning
- Metal enclosure caused some signal interference (lesson learned!)

Simple Monophonic FM Transmitter June 2013 / *Analog Electronics, Hardware*

A simple, battery-powered monophonic FM transmitter built "deadbug-style" with discrete components, designed to broadcast audio from a device like a guitar to any standard FM radio.

(Here, a photo of your deadbug-style circuit in the cigar box would be perfect!)

The Story

One of my earlier projects was this simple FM transmitter, built "deadbug-style" based on a popular online schematic. The frequency can be finely tuned with a variable capacitor, though finding the base frequency the first time was a challenge. The entire circuit, which runs off a 9V battery, is housed in a metal cigar box. Audio input is handled through a 6.3mm to 3.5mm jack adapter, making it easy to plug in almost any device. My favorite use was connecting a guitar and "playing" directly through a nearby radio.

Key Features

- Broadcasts monophonic audio over the standard FM band.
- Fine frequency tuning via a variable capacitor.
- Hand-wound inductor coil.
- 6.3mm audio input.
- Portable, self-contained unit powered by a 9V battery.

Tech Stack

- **Style:** Deadbug-style point-to-point soldering
- **Components:** Discrete transistors, resistors, capacitors (including a variable capacitor), hand-wound inductor.
- **Power:** 9V battery

Challenges & Lessons Learned

The biggest lesson from this project was in RF (Radio Frequency) design, specifically regarding enclosure materials. Housing the circuit in a metal cigar box created a Faraday cage effect, which significantly interfered with the signal strength. The transmitter only worked properly with the lid open, teaching me a valuable, practical lesson about how enclosure choices can impact antenna performance and signal propagation.

Links

- [Original Schematic/Design](#)